

Total = 12 points

Please show all your work.

1. (3 pts) Solve for x:

$$-2.7x + 4 + 3.4x = 5.6x - 16.5$$

$$4 + 16.5 = 5.6x - 3.4x + 2.7x$$

$$20.5 = (5.6 - 3.4 + 2.7)x = 4.9x$$

$$x = 20.5 / 4.9 \approx 4.18$$

Answer 4.18

2. (3 pts) Solve for t:

$$\frac{5t-8}{5} = -\frac{4t}{3}$$

$$3(5t-8) = -4t \times 5$$

$$15t - 24 = -20t$$

$$15t + 20t = 24$$

$$35t = 24$$

$$t = 24/35 \approx 0.69$$

Answer 0.693. (3 pts) If $a = -3.27$, $b = 1.5$, $c = -0.7$, find:

$$\frac{4b - 2c\sqrt{2a^3 + 18b^5} - 5}{c - 4a}$$

$$\frac{6 + 1.4\sqrt{-69.93 + 136.69} - 5}{-0.7 + 4 \times 3.27} = \frac{6 + 1.4 \times 8.17 - 5}{12.38} \approx 1.005$$

Answer 1.0054. (3 pts) Find the slope, and the x and y intercepts for the line $7x - 4y = 12$.

$$4y = 7x - 12$$

$$y = \frac{7}{4}x - 3$$

$$y=0 \Rightarrow 7x=12 \Rightarrow x=12/7$$

slope 7/4

$$x=0 \Rightarrow -4y=12 \Rightarrow y=-3$$

x-intercept 12/7y-intercept -3

$$\text{Or } 7x - 4y = 12 \Rightarrow 3y = 12 \Rightarrow y = 4$$

slope 0

x intercept 4

y intercept NONE